System 129 (2025) 103599

Contents lists available at ScienceDirect

System

journal homepage: www.elsevier.com/locate/system

Review

Beyond ideologies of nativeness in the intelligibility principle for L2 English pronunciation: A corpus-supported review

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ARTICLE INFO

Keywords:
Ideologies of nativeness
Intelligibility principle
English language teaching (ELT)
Mutual responsibility for intelligibility
Comprehensibility
Equitable L2 pronunciation teaching and research

ABSTRACT

The intelligibility principle for second language (L2) pronunciation challenges the nativeness principle that suggests that L2 speakers' goal should be nativelike pronunciation. However, in this corpus-supported review of studies based on the intelligibility principle, we show that the understanding of the principle is still underpinned by what we term ideologies of nativeness, which favor speakers of privileged first language (L1) varieties and undermine other L1 and L2 World Englishes speakers. We focus on discourses surrounding the two most prominent keywords: *errors*, frequently used to describe proficient L2 speech and blame it for compromised intelligibility, and *comprehensibility*, often uncritically used to gauge listeners' understanding although it measures subjective perception and can be closely associated with nativeness. Such discourses can obscure a more complete understanding of communication that acknowledges variation in L1 speech and the mutual nature of communication, as well as contribute toward negative perceptions of L2 speakers. Based on our review, we suggest that teachers avoid prescribed pronunciation models, instead helping learners develop their pronunciation based on a broad perceptual repertoire. We further recommend that researchers focus on how communication difficulties are successfully repaired without relying on prescribed pronunciation norms, recognizing mutual responsibility and accountability for intelligibility by all interlocutors.

1. Introduction

While globalized English allows people in all different parts of the world to communicate with each other, it also represents a source of linguistic injustice (Soler & Morales-Gálvez, 2022). English-related linguistic injustice is often connected to what we call *ideologies of nativeness*, socially shared belief systems that privilege so called native speaker varieties (discussed further in the following section). Ideologies of nativeness are pervasive across language research fields including second language acquisition research (SLA) and English language teaching (ELT), as observed by numerous researchers (e.g., Birkeland et al., 2024; Holliday, 2006, 2018; Firth & Wagner, 1997/2007; Kubota & Lin, 2006). Given that these ideologies are associated with detrimental effects on society and human relationships, a radical move away from them has been recommended (Holliday, 2006, 2018; Ortega, 2013).

With (slowly) growing resistance to ideologies of nativeness within the fields of SLA and ELT (e.g. Cook, 1999; the Douglas Fir Group, 2016), the *intelligibility principle* for teaching and researching English pronunciation (Levis, 2005, 2006, 2020; Munro & Derwing, 1995, 2020) has received recognition for challenging the deeply entrenched belief that nativeness should be the goal of

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https://doi.org/10.1016/j.system.2025.103599
Received 8 June 2024; Received in revised form 7 January 2025; Accepted 11 January 2025
Available online 17 January 2025
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second language pronunciation. Nevertheless, we argue in this paper that the intelligibility principle as it has been understood and implemented is still largely underpinned by ideologies of nativeness without fundamentally problematizing them. This is mainly due to the continued framing of English as a second language (ESL) or foreign language (EFL) for L2 learners and speakers whose pronunciation it examines, thereby evaluating their speech in reference to native speakers. We suggest that truly overcoming the reliance on native speaker models in L2 pronunciation research and ELT requires more innovative approaches.

In the following sections we define ideologies of nativeness, then review the intelligibility principle, its implementation, and previous critiques of it. We then critically engage in a corpus-supported review of published studies based on the intelligibility principle that make recommendations for pronunciation teaching (hereafter, intelligibility studies), to show in what ways and to what extent ideologies of nativeness are present and to discuss potential consequences of such ideologies. Finally, we make recommendations for working with pronunciation in SLA research and ELT without involving native norms.

2. Ideologies of nativeness

Ideologies of nativeness in this paper is a collective term for variously named but similar belief systems, including *the ideology of nativeness* (Shuck, 2006; Subtirelu, 2015), *monolingual bias* (Ortega, 2013), *monolithic views* (Firth & Wagner, 1997/2007), *native-speakerism* (Holliday, 2006, 2018), *L1 speaker norms* (Lindemann & Subtirelu, 2013), and *native-speaker norms* (Sung, 2016). The core of these ideologies is the belief in the legitimacy and superiority of native speakers over non-native speakers (Holliday, 2006, 2018; Ortega, 2013).

Whether overtly stated or subconsciously applied, ideologies of nativeness associate English with certain nations, as well as with race, ethnicity, social class, and political power (Holliday, 2006, 2018; Kubota & Lin, 2006; Shuck, 2006). Nativeness in English is typically ascribed to people born to white middle-class families in the UK or countries where English became dominant through colonization, such as Canada, the US, or Australia (Jenkins, 2015). Thus, a native speaker of English is typically imagined as a white monolingual with 'no accent' (Shuck, 2006), rather than anyone who has learned and used English from birth. Even within Anglophone countries, people with stigmatized accents are often regarded as deviating from idealized nativeness (see, e.g., Davies, 2004). Similarly, English varieties spoken in what Kachru (1992) called the Outer Circle, like India or Singapore, although commonly spoken from infancy, are often not considered native varieties but *nativized* ones, frequently labeled as "deficient and unacceptable by virtue of the local characteristics they have acquired in the process of being transplanted" (Jenkins, 2015, p. 46).

In this way, ideologies of nativeness can affect anyone whose linguistic features differ from preferred native varieties (Bamgbosç, 1998; Rajadurai, 2007), undermining not only L2 speakers but also speakers of numerous L1 varieties (Peterson, 2019). Ideologies of nativeness substantially involve beliefs and attitudes that expose those excluded from native speaker status to stereotyping, stigmatization, and discrimination in various social contexts (e.g. Kubota & Lin, 2006; Lippi-Green, 2012; Shuck, 2006; Subtirelu, 2015; Tan et al., 2021).

Unfortunately, despite some scholars' resistance (e.g. Cook, 1999; Firth & Wagner, 1997/2007; Ortega, 2013), SLA and ELT are major instruments through which ideologies of nativeness have been elaborated and perpetuated in public space, gaining authority as scientific fact or common sense (as noted by e.g., Ahan et al., 2020; Firth & Wagner, 1997/2007; Holiday, 2006; 2018). Native speakers of English are often unproblematically imagined to be the norm-providers and benchmark for learning the language (Ortega, 2013; Firth & Wagner, 1997/2007; Holliday, 2006, 2018). This emphasis on nativeness is based on the invalid assumption that monolingualism is the natural state, against which learning and using two or more languages is scrutinized and evaluated (Ortega, 2013). English L2 speakers' bilingual features differing from idealized native speakers—regardless of whether they enable the learners to be successful in using the language—are viewed as defective, problematic, insufficient, or incomplete (Firth & Wagner, 1997/2007).

3. The intelligibility principle

The intelligibility principle has been recognized as a sound alternative approach to pronunciation in teaching English as a second (ESL) or foreign language (EFL), where ideologies of nativeness are dominant, with its central argument that "learners simply need to be understandable" (Levis, 2005, p. 370). The principle challenges the nativeness principle, which denotes that "it is both possible and desirable to achieve native-like pronunciation in a foreign language" (Levis, 2005, p. 370).

Research based on the intelligibility principle operationalizes three core constructs – *intelligibility*, *comprehensibility*, and *accentedness* (Derwing & Munro, 2015; Munro & Derwing, 1995, 2020) – to evaluate L2 speakers' pronunciation. Accentedness measures the degree of difference from a native accent and is therefore mainly used for comparison with the nativeness principle. Intelligibility refers to actual understanding of a speaker's intended message (Derwing & Munro, 2015), often measured by orthographic transcriptions or comprehension questions. Comprehensibility refers to the degree of effort that listeners need to make to understand the speaker, typically measured through their subjective ratings. Whether regarded as two distinct constructs (e.g., Crowther & Isbell, 2024; Derwing & Munro, 2015), or as "not usually distinguished" from each other (e.g., Levis, 2005, 2006, p. 252; 2020; Isaacs & Trofimovich, 2012), a common proposal suggests that intelligibility and comprehensibility are closely related while both are only partially correlated with accentedness. Based on this suggestion, findings on comprehensibility have been used to suggest implications for how an L2 speaker can achieve intelligible speech (e.g., Alnafisah et al., 2022; Crowther, 2020; Crowther et al., 2015, 2018; Isaacs et al., 2018; Isaacs & Trofimovich, 2012; Munro & Derwing, 1995, 2006; Nagle et al., 2022; Nagle et al., 2023; Saito et al., 2016; Saito & Saito, 2017).

We observe several signs of intelligibility studies' attempts to eschew the nativeness principle. Centrally, their emphasis on intelligibility and comprehensibility, which they suggest only partially correlate with accentedness (which measures nativelikeness), is

to imply that the speaker can be well understood while maintaining their own accent (Crowther & Isbell, 2024; Derwing & Munro, 2015; Levis, 2005, 2020; Munro & Derwing, 1995, 2020). Some of these studies also showed that non-native speakers can be as qualified as native speakers to teach pronunciation, for example by showing that there was no significant difference between the effects of native and non-native English teachers on their learners' comprehensibility for native listeners (Levis et al., 2016). Similarly, Lascotte and Tarone (2022) and Murphy (2014) demonstrated that English learners could use L2 speakers as pronunciation models. Moreover, the foundational researchers Derwing and Munro have discussed concerns about discrimination against L2 speakers in recruitment and workplace contexts and so-called accent reduction businesses (e.g., Derwing & Munro, 2015).

These signs show that intelligibility studies have addressed some critical aspects of ideologies of nativeness, questioning the ecological validity of having nativeness as the goal for second language learning. Moreover, methodologies have evolved since Munro and Derwing's (1995) ground-breaking study, the scope of intelligibility has expanded, and researchers have debated the definitions of key constructs (Crowther & Isbell, 2024; Levis, 2020; Munro & Derwing, 2020). Nevertheless, some ideological inclinations that may be congruent with ideologies of nativeness have persisted over time across different intelligibility studies, particularly in terms of evaluating L2 speech against a limited set of 'native' varieties of English.

For example, Rajadurai (2007) problematized intelligibility studies for their bias towards native speaker norms, which often results in the description of non-native speech in terms of "errors," "deviance," "distortion," and "strange pronunciation" (p. 93). According to Rajadurai, the bias includes myths surrounding Western native varieties, which are upheld as the standard for all English speakers, with their speakers implied to be always intelligible and inherently capable of judging speaker intelligibility, ultimately attributing communication challenges largely to non-Western World Englishes speakers.

In addition, Lindemann and Subtirelu (2013) noted that intelligibility studies have focused largely on changing speakers' pronunciation to be more accurately and comfortably understood by listeners. They observed that those listeners' roles have less often been scrutinized, and even when they have, such consideration was seldom incorporated into study designs or interpretation of results (though they discussed Zielinski, 2008 as a notable exception). Lindemann and Subtirelu argued that taking listeners' comprehension of someone's speech as a straightforward measure of that speaker's pronunciation can be problematic given that biases against the speaker's variety may be widely shared, regardless of listeners' language backgrounds as L1, L2, or even speakers of the same variety.

4. Ideologies of nativeness in intelligibility studies: corpus-supported review

While intelligibility studies are recognized for providing a sound alternative to the nativeness principle, the issues noted above may be connected to what we term ideologies of nativeness, which can impede linguistic justice and equitable L2 research and teaching. Given the great influence of the earliest and subsequent intelligibility research on contemporary L2 pronunciation research and ELT, we conducted a systematic investigation to examine how such ideologies may be present in commonly shared discourses of those studies over time. That is, our study does not investigate how intelligibility research has developed or changed over time, but rather examines the discourses that are repeated up to the present despite the methodological and theoretical evolution of the field. Our intention thereby is not to criticize individual researchers, who, like ourselves, may consciously disagree with such ideologies but may not see practical ways to extricate themselves from the ideologies' deep permeation of the field. Rather, our intention is to help make ideologies of nativeness in certain taken-for-granted constructs more visible and to propose alternatives that can move us still further towards a more accurate and equitable view of pronunciation.

To that end, we adopted corpus-supported discourse analysis (Lee, 2008) involving *keyness* analysis that examined highly prominent words—keywords—in intelligibility studies. Keywords in a set of texts, or corpus, reveal what is important in these texts and what they are "really about" (Scott & Tribble, 2006; Baker, 2006). Keywords are identified by comparing this target corpus with a reference corpus that shares important features but contrasts in particular points, where the relative size of both corpora and the relative frequencies of each word in the corpora are considered (Scott & Tribble, 2006; Baker, 2006). Identified keywords are then examined in context to observe discourse patterns and how such discourses may reproduce and perpetuate certain ideologies (Baker, 2006).

Following these principles, we compiled two corpora as described in our online supplemental material that also provides bibliographic information of the studies in the two corpora and keyness analysis procedures. The target corpus comprised 49 articles published between 1995 and 2023 that framed English as a second or foreign language for L2 users and learners, evaluating their speech in reference to native speakers when operationalizing intelligibility and/or comprehensibility for implementing the intelligibility principle. Studies that operationalized comprehensibility without linking it to the intelligibility principle were excluded. The starting year (1995) was chosen because it marks the publication of the foundational intelligibility study by Munro and Derwing (1995). Although this study was published 30 years ago, it continues to be highly cited by contemporary studies, and a recently published redux article (Munro & Derwing, 2020, not included in the corpus due to its substantial textual overlap with the 1995 paper) has renewed its relevance. Likewise, other early publications have been consistently cited by subsequent studies to support their arguments, through which certain ideologies are maintained and reproduced over time.

Since ideologies of nativeness among intelligibility studies appear to be closely linked to how these studies frame English and its learners and users, we compiled in the reference corpus 48 articles published between 1995 and 2023 that contrast with target corpus studies in their framing of English and its users and learners, while sharing similar topics, textual features and readership. Specifically, the articles in the reference corpus investigated the intelligibility of pronunciation from the perspectives of English as a lingua franca (ELF), English as an international language (EIL), or World Englishes (WE), challenging the practice of evaluating L2 speech in comparison to native speakers.

The ideological distinction between the target and the reference corpora is nevertheless by no means absolute. A few researchers

have published in both paradigms, leading to some of their studies being classified into the target corpus and others in the reference corpus. Most ELF/EIL/WE-oriented pronunciation studies in the reference corpus frequently drew on major intelligibility studies. As discussed in Section 4.3, reliance on British or American English norms for intelligibility is still observed in reference corpus studies, although they generally described the features of different English varieties descriptively rather than evaluating them against so-called native speaker norms.

For keyness analysis, we used AntConc 4.2.0 (Anthony, 2022), involving sets of log-likelihood. The log-likelihood was based on contingency tables that compared the observed and expected values of a target word with those of all other words in both the target and reference corpora (Anthony, 2023). Normality is not required in log-likelihood-based keyness analysis; the essential assumptions are the independence of observations in the target and reference corpora and a sufficiently large sample size for investigation (Dunning, 1993). The *p*-value is loosely interpreted as a basis for identifying and ranking keywords, not for rejecting a null hypothesis (Pojanapunya & Todd, 2018). Baker (2006, p. 125) notes, "the smaller the p-value, the more likely it is that the word's strong presence in one of the sub-corpora isn't due to chance but is a result of the author's (conscious and subconscious) choice to use that word repeatedly." Therefore, we selected in AntConc the highest log-likelihood threshold of 15.13, corresponding to the lowest alpha level of 0.0001. Bonferroni corrections for each test are noted in the relevant tables; the log-likelihood of all words presented in the tables is statistically significant.

Table 1 shows the top ten keywords, that is, the most prominent words in the target corpus in relation to the reference corpus. The following sections examine discourses around the two top keywords *errors* and *comprehensibility*, during which other keywords in the table are discussed. Extracts are also provided to exemplify particular points; they were identified through first looking at the concordances of keywords, and then checking the source articles. All underlining in extracts is our emphasis.

4.1. Ideologies of nativeness in discourses about 'errors'

The highest-ranked keyword in Table 1 is *errors*, found in articles throughout the entire period of the corpus. This ranking does not include the singular form *error*, which AntConc counted separately, and which appears an additional 276 times in 32 out of 49 intelligibility studies articles, not including usages like 'error bars' in statistical analysis. *Errors* is also the twelfth-most common content word in the intelligibility studies corpus overall. As seen in Table 2, two-word clusters with *errors* on the right showed that *errors* was used mostly to refer to the pronunciation features and sometimes grammar features of multilingual L2 speakers (usually identified as non-native speakers in the studies), although it also occasionally referred to listeners' transcriptions that did not match what the speakers said. While the word *errors* was widely dispersed in 88% of the intelligibility studies articles (plus one study that used *error* but not *errors*), it was especially frequent in papers investigating the role of functional load.

4.1.1. Defining language features as errors when they differ from prestigious L1-speaker features

In identifying features as errors, intelligibility studies did not consider whether the speaker was actually intelligible or highly proficient. In fact, the speakers who provided samples of 'errors' were often described as competent in English, as seen in extracts 1–3.

1) All were proficient speakers of English who had scored no less than 550 on the TOEFL … [and who had] spent a minimum of one year in Canada as graduate students … (Munro & Derwing, 1995, p. 78; also in the redux article, Munro & Derwing, 2020, p. 286, not included in the target corpus)
2) As can be seen in Figures 1 and 2, high-proficiency speakers … still produce errors in their pronunciation, but Cambridge ESOL raters considered their speech to be highly proficient (Kang & Moran, 2014, p. 184).

Table 1
Top ten keywords of intelligibility studies articles omitting the name, 'Derwing'. Frequencies are normalized per 100,000 words, with raw numbers in parentheses. Range = percentage (and raw number) of texts.

Type	Rank	Target corpus (339,298 tokens, 49 texts)				Reference corpus (320,267 tokens, 48 texts)				Keyness[b]
		Frequency		Range		Frequency		Range		
errors[a]	1	283	(959)	88%	(43)	22	(72)	46%	(22)	857.9
comprehensibility[a]	2	589	(1999)	86%	(42)	204	(653)	63%	(30)	643.0
L (language)	3	875	(2968)	98%	(48)	397	(1272)	94%	(45)	607.2
speech	4	693	(2350)	100%	(49)	293	(937)	98%	(47)	551.5
instruction	5	149	(505)	84%	(41)	69	(221)	29%	(14)	358.1
accentedness	6	248	(843)	82%	(40)	21	(66)	40%	(19)	353.3
judgments	7	110	(373)	76%	(37)	12	(40)	19%	(9)	291.1
Fluency	8	141	(477)	57%	(28)	32	(101)	31%	(15)	244.6
Rating	9	136	(460)	90%	(44)	32	(103)	21%	(10)	224.7
Stimuli	11	71	(240)	55%	(27)	5	(17)	19%	(9)	213.6

Note.
[a] The absolute ranks of the words *errors* and *comprehensibility* in the target corpus (excluding function words) were 12th and 3rd, respectively.
[b] Log-likelihood alpha = 0.0001; the new alpha adjusted through Bonferroni correction = $2.1 * 10^{-7}$.

Table 2
2-word clusters containing errors on right with minimum frequency 10, minimum range 5 (at least 10% of texts). All examples of load errors are part of the three-word cluster functional load errors. Frequencies and ranges are normalized per 100,000 words, with raw numbers in parentheses. Range = percentage (and raw number) of texts.

Cluster	Rank	Frequency		Range	
segmental errors	2	33	(112)	51%	(25)
of errors	3	19	(64)	35%	(17)
stress errors	4	10	(34)	29%	(14)
pronunciation errors	5	10	(33)	35%	(17)
grammatical errors	7	7	(24)	12%	(6)
(functional) load errors	8	7	(23)	10%	(5)
the errors	10	6	(19)	22%	(11)
phonemic errors	12	5	(17)	10%	(5)
phonological errors	14	4	(12)	10%	(5)
consonant errors	15	3	(11)	10%	(5)

3) All were graduate students … at a Midwestern Research-1 university in the United States … at an advanced level of English … many such individuals serve as teaching assistants in basic disciplinary courses, necessitating that their speech be intelligible and comprehensible (Alnafisah et al., 2022, p. 4).

In other words, the speech of proficient, intelligible bilingual users was described in terms of errors and in some cases used to decide which pronunciations were considered problematic. Although Kang and Moran (2014, p. 184) did not assume the pronunciations were problematic, even noting that they "appear across varieties of English worldwide", the study still labeled them as errors. Such labels negatively characterize L2 features in comparison to idealized native speaker features, reflecting ideologies of nativeness.

Intelligibility studies argue that some errors do not impact intelligibility and not all L2 speaker features that they identify as errors equally affect intelligibility, as seen in extract 2 above, as well as extracts 4–6. This is in fact the key contrast between the intelligibility principle and the nativeness principle. For example, applying the concept of *functional load* (FL), a measure of how much work is done by two contrasting phonemes for meaning discrimination (Catford, 1987), studies distinguished between high FL errors and low FL errors to suggest that the former are more detrimental to intelligibility than the latter (Alnafisah et al., 2022; Munro & Derwing, 2006).

4) Munro (1998) proposes the concept of "error gravity'', and notes that some L2 pronunciation errors are more serious than others, since they could affect speech intelligibility (Zhi & Li, 2021, p. 3).
5) Indeed, high functional load (FL) errors caused a greater reduction in comprehensibility ratings than did low functional load errors. (Derwing & Munro 2009b, p. 482)
6) According to (Levis, 2006), some pronunciation errors are more likely to affect the intelligibility and comprehensibility of speech (e.g., the distinction between/l/and/r/) while others are likely to affect only whether speech is heard as accented (e.g., the distinction between/θ/and/f/, as in thought-fought) (Alnafisah et al., 2022, p. 18)

As such, a major concentration in the discourse of errors reflects the endeavor to reduce the burden on the learner imposed by the nativeness principle that treats all non-native features as needing to be changed, making the learning goal more achievable. However, the word *error* inherently means something considered "wrong", "not correct", "not accurate" or "not giving the right result" (Cambridge Dictionary, n.d. Error). Calling an English user's features *errors* whenever they are perceived to be non-native reifies the naturalized ontological constructs entrenched in SLA and ELT of ideal native speakers and ever-deficient non-native speakers (Firth & Wagner, 1997/2007; Holliday, 2006, 2018; Ortega, 2013). Critically, this labeling of a person's clearly intelligible speech as erroneous reinforces and provides academic support for the societally widespread biases against all varieties not perceived as native (Lippi-Green, 2012), working against intelligibility researchers' attempt to counteract such negative social consequences.

4.1.2. (Still) considering L2 features the (main) reason for compromised intelligibility

Intelligibility is by right mutually accountable among interlocutors, as it refers to "the degree of match between a speaker's intended message and the listener's comprehension" (Derwing & Munro, 2015, p. 5). However, intelligibility studies have still typically presented the L2 speaker's features, which they call *errors*, as the main cause of compromising intelligibility (Rajadurai, 2007), suggesting an underlying ideology of nativeness. It indicates an implicit assumption that so-called native speakers do not make errors, are automatically intelligible, and thus are not accountable at all *as speakers* for intelligibility, although some suggest their responsibility *as listeners* to make efforts to understand L2 speakers. This assumption can be seen in statements suggesting that intelligibility is directly and sometimes even uniquely tied to L2 speech features differing from idealized native speech, in some cases with *errors* as a grammatical subject causing communication difficulties, as already seen in extracts 5 and 6, as well as in extracts 7–9.

7) Our finding converges with language researchers' claim (e.g. Anderson-Hsieh et al., 1992; Gilbert, 1995; Celce-Murcia et al., 1996; Munro & Derwing, 1997; Mouri, Hirose, & Minematsu, 2003) that prosodic errors seriously compromise speech intelligibility (Yenkimaleki & van Heuven, 2019, p. 302).

8) In alignment with previous research (Bent et al., 2007; Zielinski, 2008), vowel errors were found to reduce the intelligibility of speech (Wheeler & Saito, 2022, p. 12).
9) For example, Crowther, Trofimovich, Saito, and Isaacs (2015b) confirmed that … Whereas segmental errors were a barrier to English comprehensibility in the case of L1 Chinese speakers, for L1 Hindi-Urdu speakers, pronunciation fed into listeners' perceptions of how accented they sounded but did not impede their comprehensibility (Isaacs et al., 2018, p. 200).

The assumption that the L2 speaker's pronunciation is at fault for communication problems is understandable to some extent, primarily given the concentration of intelligibility studies in Western Anglophone countries like Canada or the US, where pronunciation instruction often addresses the challenges faced by local listeners. However, even within these countries, English accents vary widely, including both L1 and L2 speakers'. Encouraging learners to appropriate pronunciation features that are closely associated with privileged white middle-class speakers may therefore contribute to perpetuating social inequality and accent-based discrimination. For example, language learners who are taught according to such models can become highly intolerant of the other varieties they hear in their daily life, becoming upset with L1 and L2 speakers who do not speak 'correctly'.

Moreover, identifying L2 features as errors based on idealized native norms is even more problematic in ELF settings, where a great diversity of English pronunciation is commonplace, as noted by Matsumoto (2011). In such settings, promoting a few Western varieties' pronunciation norms for securing intelligibility can undermine speakers of all other varieties, particularly when their phonology largely differs from the promoted pronunciation. Moreover, norms from these Western varieties are often irrelevant to ELF communication given that speakers of those varieties are often absent, and their features might greatly differ from the actual participants' features (Jenkins, 2000). Although prestigious L1 speaker features often score high on tests of intelligibility and comprehensibility probably due to their status and in some cases greater familiarity, other varieties, even less familiar ones, have sometimes been found to be equally or even more intelligible to listeners in ELF contexts (Hansen Edwards et al., 2018; Matsuura, 2007; Smith & Nelson, 2019).

4.2. Ideologies of nativeness in discourses about comprehensibility

The second most frequent keyword and the third most frequent content word after *speech* and *listeners* in the target corpus is *comprehensibility*, typically measured by listeners' subjective perception of the effort they make to understand a speaker (Derwing & Munro, 2015). The heavy reliance on this construct in studies based on the intelligibility principle raises additional concerns discussed in this section. The additional keywords *judgments* and *ratings* in Table 1 reflect the experimental orientation of many intelligibility studies that focus on comprehensibility, which often grant listeners authority in determining the functionality of speakers' pronunciation. Like the focus on L2 speaker *errors*, the use of comprehensibility reflects the overemphasis on the responsibility of the speaker for comprehension. Listeners' ratings are taken as a direct measure of speaker problems, without acknowledging ideological components of these judgments (Lindemann & Subtirelu, 2013).

As seen in Table 3, the most probable collocate of comprehensibility is *accentedness* (although observed in a narrower range of papers), while *intelligibility* ranks only as the 8th most probable collocate by log-likelihood. This reflects the fact that many studies promoting the intelligibility principle operationalized only comprehensibility and accentedness, highlighting the value of the concept of comprehensibility for speech evaluation and for proposing implications for pronunciation teaching (e.g., Alnafisah et al., 2022; Crowther et al., 2015, 2018; Isaacs & Trofimovich, 2012; Kennedy & Trofimovich, 2019; Munro & Derwing, 2006).

4.2.1. Promoting comprehensibility for implementing the intelligibility principle, despite its close relationship with accentedness (nativelikeness)

As discussed earlier, studies advocating for the use of comprehensibility in implementing the intelligibility principle emphasizes its alignment with intelligibility, contrasting it with accentedness, which is regarded as aligned with nativeness. (e.g., Crowther, 2020; Crowther & Isbell, 2024; Isaacs & Trofimovich, 2012; Levis, 2006, 2020; Munro & Derwing, 1995, 2006; Nagle et al., 2022; Nagle et al., 2023). Reflecting this discourse, over half of the studies in the target corpus, spread over the entire review period, either

Table 3
Collocates of comprehensibility with high log-likelihood (the probability of co-appearance of the target word and a collocate), within 5 spaces to both left and right with minimum frequency 10, minimum range 5 (at least 10% of texts). Frequencies and ranges are normalized per 100,000 words, with raw numbers in parentheses. Range = percentage (and raw number) of texts.

Collocates	Rank	Log-likelihood*	Frequency		Range	
accentedness	1	1212.6	133	(450)	67%	(32)
ratings	2	620.1	88	(300)	65%	(31)
and	3	383.2	309	(1048)	81%	(39)
judgments	4	268.8	40	(135)	44%	(21)
perceived	5	217.7	29	(100)	29%	(14)
fluency	6	200.8	39	(131)	23%	(11)
accent	7	158.0	46	(157)	48%	(23)
intelligibility	8	134.0	62	(210)	79%	(38)
scores	9	87.0	29	(97)	42%	(20)

Note. *Log-likelihood alpha = 0.0001; the new alpha adjusted through Bonferroni correction = 0.0000062.

operationalized only comprehensibility or conflated it with intelligibility in implementing the intelligibility principle. Justifications for operationalizing only comprehensibility varied, however. Some studies explicitly or implicitly treated intelligibility and comprehensibility as broadly the same construct for measuring understanding (e.g., Alnafisah et al., 2022; Derwing and Munro, 2009a; Gordon & Darcy, 2022; Saito & Saito, 2017), while others argued that comprehensibility is an effective and practical tool for measuring listeners' understanding easily and quickly, closely reflecting intelligible pronunciation in real-life interactions (e.g., Crowther, 2020; Crowther et al., 2015, 2018; Foote & Trofimovich, 2018; Isaacs & Trofimovich, 2012; Suzuki & Kormos, 2020).

However, correlations suggest that comprehensibility is less closely related to intelligibility, despite the prevailing discourse that they both "as measure of understanding" are "often moderately-to-strongly correlated" (e.g., Crowther & Isbell, 2024 p. 2; Derwing & Munro, 2015). Lindemann and Subtirelu (2013) reviewed the five studies that had reported correlations between intelligibility (in their terms, "objective I/C") scores and comprehensibility ("subjective I/C"). Two of the studies (Munro & Derwing, 1995; Derwing & Munro, 1997) with Canadian English L1 listeners reported moderate (~0.5) correlations on average, but not all listeners showed this trend. The remaining three studies, which used L2 listeners, showed weak to nonexistent correlations (0.34–0). Similarly, a recent study with Swedish listeners (Jeong, Elgemark, & Thorén, 2021) found no correlation between intelligibility and comprehensibility.

In contrast, comprehensibility was sometimes strongly correlated with accentedness, suggesting greater alignment with the nativeness principle than the intelligibility principle. The two studies using L1 listeners mentioned above (Munro & Derwing, 1995; Derwing & Munro, 1997) found correlations with accentedness similar to those with intelligibility (0.45–0.6), while more recent studies with North American English L1 listeners reported even stronger correlations between comprehensibility and accentedness: Kang (2010) found 0.61; Saito, Trofimovich, and Isaacs (2016) found 0.89; and Crowther et al. (2018) found 0.74 to 0.80. Thus, while correlations vary depending on speakers, listeners, and tasks, there seems to be no solid ground for considering comprehensibility as aligned more with intelligibility than with accentedness. Given that accentedness is generally understood as referring to nativeness, emphasizing comprehensibility therefore may result in reverting to the nativeness principle, as we discuss further in the following section.

4.2.2. Uncritical of signs of nativeness in listeners' comprehensibility ratings

Several intelligibility studies reported findings that point to nativeness in listeners' comprehensibility ratings, largely without critical discussion. These studies investigated not only pronunciation but also other aspects of L2 speech, including speech rate, lexical appropriateness, grammatical accuracy, and sophistication and richness in discourse (e.g., Crowther et al., 2015, 2018; Isaacs et al., 2018; Isaacs & Trofimovich, 2012; Saito et al., 2016; Saito & Saito, 2017). They have found close associations between listeners' comprehensibility ratings and different linguistic aspects, although the associations can vary in relation to other factors such as speakers' L1s.

Crucially, these findings suggest that comprehensibility ratings may in fact reflect the degree of nativeness across various linguistic aspects. Specifically, many of the factors associated with comprehensibility, such as accuracy in grammar, word stress, and rhythm, as well as appropriate lexical choice judged by raters using native speaker norm-based criteria, can be expected to assess nativeness. Crowther et al. (2018) even reported that the linguistic distinctions between ratings of comprehensibility and accentedness (which measures nativelikeness) tended to blur when L2 speakers engaged in complex tasks, where the two constructs had similar associations with various linguistic dimensions.

In fact, as seen in extracts 10 and 11, the findings about the relationship between comprehensibility and different linguistic dimensions sometimes led to pedagogical suggestions that may be uncritically congruent with the nativeness principle, rather than the intelligibility principle:

10) While an emphasis on segmental accuracy and grammatical complexity plays an important role in accentedness (especially at high ability levels), <u>a tailored approach is needed for L2 comprehensibility, with prosody, temporal variables, and lexical accuracy ideally targeted for beginner to intermediate learners, and segments, prosody, and grammatical accuracy for intermediate to advanced learners</u> (Saito et al., 2016, p. 234).
11) In terms of pedagogical implications, the finding that linguistic distinctions between accentedness and comprehensibility are blurred in a complex task might be encouraging for both language learners and teachers. <u>Using complex tasks, L2 speakers might be able to practice a range of linguistic targets (phonology, fluency, lexis, and grammar), with most having a bearing on both accentedness and comprehensibility</u> (Crowther et al., 2018 p. 455).

While discussions like the ones in extracts 10 and 11 may intend to offer a choice of which features to change, there is the danger that they could be interpreted and applied as suggesting that learners need to master them all. In that interpretation, the pedagogical recommendations seem unconnected to an intelligibility principle that "holds that learners simply need to be understandable" (Levis, 2005, p. 370) or to prioritizing a limited number of linguistic features to make L2 learning more practical and achievable, as many intelligibility studies argue for. Moreover, given that the raters who evaluated L2 speech were based on native speaker norm-based criteria, the recommended linguistic targets are likely equivalent to native speaker norms. A reasonable conclusion can therefore be that based on their findings, such studies indirectly encouraged learners to work towards nativelikeness in several linguistic dimensions.

In sum, correlational data and in some cases pedagogical recommendations from these studies do not match the common justification for using the construct *comprehensibility* to implement the intelligibility principle. Crucially, these findings suggest that listeners' ratings of comprehensibility are likely linked to the speakers' nativeness. Yet, pedagogical discussions operationalized comprehensibility based on those ratings, reflecting an ideology that evaluates L2 speech features against native speaker norms or L1

speaker perceptions. However, simply adding L2 listeners or examining interaction among L2 interlocutors without scrutinizing their underlying accent attitudes in comprehensibility ratings would not address the problem, as we discuss in Section 5.1 below.

4.3. Negative keywords: Suggestions for moving away from ideologies of nativeness

Similar to intelligibility studies, many ELF pronunciation studies continued to rely on UK or US pronunciation norms to suggest features deemed important for speaker intelligibility, promoting some 'standard native speaker' features as desirable (e.g., Jenkins, 2000; Jeong et al., 2020; Kang et al., 2018; Sewell, 2017; Walker, 2010). While a few ELF studies (e.g., Matsumoto, 2011; O'Neal, 2015, 2019) framed intelligibility as something interlocutors achieve contextually through the negotiation of meaning and use of repair strategies, several of them had non-interactional experimental designs. For example, based on their experiment, Jeong et al. (2020) recommended realizing three phonetic features known to be part of the UK or US standard pronunciation system for ELF communication. Many ELF pronunciation studies also assumed that the pronunciation of L2 speakers and even of speakers from well-established new varieties is the main cause of communication breakdown (e.g. Deterding, 2013; Jenkins, 2000). Moreover, a few studies in the reference corpus such as Nagle et al. (2022) operationalized comprehensibility in the context of ELF communication without critically examining the construct's close relationship with accentedness discussed in Section 4.2.

Nevertheless, a brief consideration of ELF intelligibility research can give some sense of a future direction for addressing intelligibility without a heavy reliance on idealized native speaker norms. Pointing in this direction, the negative keywords, words that are much more common in the reference corpus than the target corpus, are presented in Table 4.

The negative keywords *international*, *varieties*, and *Englishes*—not to mention *ELF* itself and the specific contexts mentioned in various papers—show a focus on English as an international communication tool or lingua franca rather than merely the language of a nation, emphasizing the plurality of English. Interestingly, *intelligibility* itself was likewise a negative keyword. While it appears substantially in all intelligibility studies, it is still much less frequent than in the reference corpus, likely because of the use of measures of comprehensibility as a substitute for intelligibility measures in many of the target corpus studies. Finally, the negative keyword *mutual* shows an important aspect of intelligibility, that it is achieved contextually through mutual effort and accountability in communication by all interlocutors rather than through acquiring prescribed L1 phonological rules.

5. Implications and suggestions for researching and teaching pronunciation

Ideologies of nativeness persist in studies advocating the intelligibility principle, despite researchers' positive intentions and overt stances against such ideologies. Our corpus-supported review demonstrates that such ideologies are embedded in the longstanding and still current understanding of the intelligibility principle, even though it has attempted to counteract the harmful consequences of these ideologies. The prominence of the words *errors* and *comprehensibility* in the target corpus reveals a bias against L2 speakers in evaluating their language features in view of idealized native speaker norms. Labeling L2 features as errors reinforces biases against many World Englishes whose linguistic features differ from a few Western L1 varieties. Moreover, we discussed emphasis on comprehensibility, closely linked to accentedness (a measure of nativelikeness), may maintain ideologies of nativeness rather than challenging them. In both cases, we problematized the assumption that ties intelligibility directly to L2 speech features as if L2 speakers are mainly responsible for communication difficulties. We alternatively considered some ELF studies' emphasis on mutual effort in communication (e.g., Kim, 2018; Matsumoto, 2011; O'Neal, 2015) as a healthier approach to intelligibility, moving beyond the ideologies of nativeness involved in relying on prescribed native speaker features.

5.1. Research

Overall, our review of intelligibility studies supports the need for a much more fundamental turn towards effectiveness-based

Table 4
Top ten negative keywords of the intelligibility studies articles, omitting names (Deterding, Jenkins), function words (is), and keywords appearing in fewer than 50% of the reference corpus papers (misunderstanding, Hong Kong, Singapore, Swedish, German, Thai). Frequencies and ranges are normalized per 100,000 words, with raw numbers in parentheses. Range = percentage (and raw number) of texts.

Type	Rank	Target corpus (339,298 tokens, 49 texts)				Reference corpus (320,267 tokens, 48 texts)				Keyness*
		Frequency		Range		Frequency		Range		
ELF	1	7	(25)	6%	(3)	231	(741)	60%	(29)	884.4
English	2	569	(1932)	100%	(49)	1227	(3931)	100%	(48)	822.9
international	3	19	(64)	53%	(26)	170	(545)	81%	(39)	463.5
varieties	4	9	(30)	25%	(12)	138	(442)	81%	(39)	455.4
LFC	5	4	(13)	8%	(4)	89	(285)	50%	(24)	322.3
lingua franca	7	7	(25)	29%	(14)	91	(290)	88%	(42)	277.7
intelligibility	9	423	(1434)	100%	(49)	725	(2323)	100%	(48)	268.3
core	14	9	(30)	37%	(18)	83	(265)	79%	(38)	228.9
mutual	23	6	(21)	22%	(11)	63	(202)	58%	(28)	180.6
Englishes	24	3	(11)	12%	(6)	53	(169)	63%	(30)	176.0

Note. *Log-likelihood alpha = 0.0001; the new alpha adjusted through Bonferroni correction = $1.8 * 10^{-7}$.

speaker evaluation that avoids recommending an approximation to idealized native speech. To this end, we urgently advocate for changes in terminology, such as using *features* rather than *errors*. Labeling L2 pronunciation features as errors when they differ from so-called native-speaker norms is inaccurate, since it implies a uniformity among native speech that does not exist (Preston, 2008) and invokes a deficit view. This practice can even penalize highly effective World Englishes speakers for not using particular pronunciation features and encourage learners to perceive these speakers' pronunciations as 'bad' or 'wrong'.

However, simply changing terminology is insufficient if L2 features are still assumed to be the cause of communication problems, rather than, for example, listeners' lack of familiarity with different varieties. Replacing harmful terminology should therefore include deeper reflection on how to discuss and address miscommunication. As discussed above, intelligibility, defined as "the degree of match" between the speaker's message and the listener's understanding (Derwing & Munro, 2015, p. 5), requires mutual responsibility by all interlocutors (Smith & Nelson, 1985, 2019). Although intelligibility studies have occasionally suggested training L1 listeners (e.g. Munro, 2011), their focus has been overwhelmingly on evaluating bilingual L2 speaker's pronunciation against listeners' perception. Here, listeners are very often L1 speakers of a few privileged English varieties. Even when L2 speakers are included as listeners, there is little discussion acknowledging that, through their English education, most L2 speakers are primarily familiar with prestigious Anglophone pronunciation and often admire 'native speaker' accents (Sung, 2016; Tsang, 2019). Therefore, addressing and challenging the fundamental bias against the L2 speaker in accountability for intelligibility is essential, going beyond simply adding L2 listeners to the study design.

Moreover, as we discussed earlier, comprehensibility may be more closely related to accentedness than to intelligibility, reflecting listeners' language ideologies. These points suggest that findings about comprehensibility should not be uncritically applied to language teaching, even when more interactional and contextually-bound approaches are taken, as suggested or done by recent studies (e.g., Crowther & Isbell, 2024; Nagle et al., 2022; Nagle et al., 2023).

While welcoming study designs that attend to interlocutors' mutual comprehensibility in interaction, we argue such approaches must still critically address the underlying language attitudes reflected in comprehensibility ratings, particularly examining whether and to what extent these attitudes are influenced by native-speaker-centric views. Investigations of intelligibility in interaction should also focus on how communication difficulties are successfully repaired without relying on prescribed pronunciation norms (as also recommended by Crowther & Isbell, 2024 and already begun by Kim, 2018; Matsumoto, 2011; O'Neal, 2015, 2019). Moreover, while thus far studies taking interactional approaches have focused only on L2-L2 interactions, their scope should expand to include L1 and L2 English users from diverse varieties, including examination of the role of L1 English speakers' pronunciation in communication difficulties. Better knowledge of L1 speaker challenges and strategies can inform interventions for L1 speakers and limit the accountability of L2 speakers, overcoming the tendency of constructing L1 interlocutor difficulties as being caused by L2 interlocutors.

Finally, research on comprehensibility and accentedness involving various linguistic dimensions (e.g., Crowther et al., 2015, 2018) can be very fruitful for providing insights into the constructs, if it recognizes their ideological nature, including how they are affected by listener attitudes, knowledge, and beliefs. Such research can provide an opportunity to consider linguistic measures alongside listener factors but should not be used to directly and uncritically suggest applications of findings to L2 pronunciation teaching and learning.

5.2. Teaching

While intelligibility studies have begun to shift the focus away from the nativeness principle, ideologies of nativeness continue to influence pronunciation teaching. Therefore, there is a clear need to rethink how pronunciation can be addressed in the English classroom.

A few intelligibility studies suggest promoting as pronunciation models L2 speakers who are deemed to be intelligible by pronunciation experts (e.g. Lascotte & Tarone, 2022; Murphy, 2014). We welcome such approaches that promote L2 speaker pronunciation models but wish to highlight the contextual nature of intelligibility. That is, rather than encouraging learners to choose an L2 model who largely conforms to the L1 norms found to be important for intelligibility in a decontextualized way, teachers can help them develop pronunciation strategies such as repairing or adjusting their pronunciation to facilitate different international interlocutors' understanding in specific contexts (Matsumoto, 2011; O'Neal, 2015, 2019). Teachers can also empower learners to explore and develop their pronunciation without involving GA, RP, or other prescribed norms, as exemplified by a teaching practice at a Swedish university (Jeong et al., 2022). The teacher provided exposure to various sound segments and word stress patterns from different English varieties in an equal, non-judgmental way, to help her students to discover and value their own pronunciation features.

The students in Jeong et al. (2022) were at upper-intermediate to advanced levels and generally did not have pronunciation-related challenges, but a similar approach can be taken even with beginner learners. For example, from a very early stage of an L2 learning, learners can listen to diverse pronunciations of the same words, sentences, or texts to develop a broad perceptual repertoire of English phonology, which they can use to explore and develop their own pronunciation. Such a broad repertoire can help teachers to decide when and how to provide corrective feedback without referring to RP or GA. For example, a teacher may choose to help a learner improve the pronunciation [paɪn] for the word *fine*, considering that the word-initial sound is consistently pronounced as [f] across a wide range of English varieties. Meanwhile, the teacher can encourage learners' autonomy in establishing their own pronunciation for aspects that vary widely among World Englishes speakers, such as the degree of aspiration for the first consonant of *point* or the shape of the vowel in *shirt*. The broad input as resources for pronunciation instruction can also allow the teacher to raise learners' awareness of less common variants in established varieties, such as the/l/-/ɹ/merger in Kenyan English or tapping intervocalic/t/in American English, as some acceptable pronunciations, without stigmatizing or conversely idealizing certain varieties' sound systems.

In fact, listening training for diverse English accents, which has been done with North American English L1 speakers (e.g., Derwing

et al., 2002; Lindemann et al., 2016), is necessary for all English users (Jeong, Elgemark, & Thorén, 2021, Ludwig & Mora, 2017; Jeong et al., under review), regardless of their linguistic background or proficiency level, as initial difficulties in understanding interlocutors' pronunciation is common in international communication (Deterding, 2013; Jenkins, 2000). An explicit emphasis is therefore needed on mutual accountability for intelligibility among both L1 and L2 users as both speaker and listener, with accommodation and repair strategies for tackling miscommunication and facilitating negotiation of meaning. To that end, both learners and lifelong users of English should be encouraged to recognize their own role as listeners in achieving intelligibility and be exposed to a diverse range of English accents, including those of successful transnational English users. Such exposure can also help challenge the tendency among learners to aspire to 'native speaker' pronunciation, often as a result of the prominence of these accents in educational materials (Sung, 2016; Tsang, 2019).

Research findings suggest that individuals can quickly adapt their perception to new accents through high variability perception training (e.g., Baese-Berk et al., 2013; Jeong et al., under review). Jeong and Lindemann (under review) particularly show the feasibility of training learners to understand accent variation and fostering positive attitudes towards diverse accents in ELT contexts. Additionally, the English-teaching repertoire can include critically addressing ideologies of nativeness and introducing successful English users from non-privileged linguistic backgrounds in the English classroom. When English users and learners are competent in understanding a wide range of World Englishes accents, the aspiration for achieving nativelike pronunciation may decrease gradually, either partially or entirely.

5.3. Conclusion

The intelligibility principle has significantly challenged the nativeness principle. The major discourses in intelligibility studies over the three decades that we reviewed were, however, still influenced by ideologies of nativeness, favoring so-called native speakers and disadvantaging L2 speakers of English. Evaluating L2 pronunciation against an idealized native speaker standard, as reflected in keywords like *errors* and *comprehensibility*, may reinforce linguistic injustice and hamper equal relationships in international communication. To address these issues in how the intelligibility principle has been applied, we call for a fundamental rethinking of the native speaker and non-native speaker dichotomy for understanding English pronunciation. This includes radical actions like avoiding using *errors* for describing L2 pronunciation, avoiding taking listeners' ratings as direct measures of speakers' abilities, and focusing on how communication difficulties are successfully repaired without relying on prescribed pronunciation norms, where intelligibility should be viewed as a mutual responsibility among all involved, irrespective of their linguacultural backgrounds. Learners should be helped to explore their pronunciation on its own terms, foster their abilities to understand a wide range of accents, and critically reflect on their own attitudes towards privileged and unprivileged accents. By implementing these actions and practices, we researchers and teachers can promote linguistic justice, ultimately moving away from the entrenched ideologies of nativeness.

Credit author statement

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Declarations of interest

None.

Acknowledgements

We are very grateful to Editor-in-Chief Dr. Yongyan Zheng and our three reviewers for their invaluable feedback. Remaining errors are our own.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.system.2025.103599.

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